



# <Library Book Borrowing System>

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## **A. PROJECT DESCRIPTION**

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Library Borrowing System is a web based library borrowing system developed to address common problems in managing library resources. Traditional methods of tracking borrowed books can be time consuming and may lead to forgotten due dates and unmonitored penalties.

This project was created to provide a simple and organized system for students to easily manage their borrowed books. It focuses on delivering a user friendly interface that allows efficient access to library features.

The system allows users to log in, browse available books, borrow and return them, track due dates, and monitor penalties. It aims to simulate a functional library system using basic web technologies.

## **B. OBJECTIVES**

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- To create a system that allows users to borrow and return books easily.
- To provide a simple way to track borrowed books and due dates.
- To help users monitor penalties for late returns.
- To design a user friendly interface for better experience.
- To demonstrate how a library system works using web technologies.

# C. SCOPE AND FEATURES

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The system is limited to a browser based application and does not require internet connection after loading. It is designed for basic library operations and is intended for students and small scale use.

## C.1 Scope and Feature #1

Features

User Login System

Users can log in using predefined accounts to access the system.

## C.1 Scope and Feature #2

Dashboard Overview

Displays user information, number of borrowed books, due dates, penalties, and member rank.

## C.1 Scope and Feature #3

Book Catalog

Users can browse available books, search by title, and filter by category.

## C.1 Scope and Feature #4

Borrowing System

Allows users to select a book, set borrow and return dates, and confirm borrowing.

## C.1 Scope and Feature #5

Due Date Monitoring

Displays borrowed books along with due dates, status, and penalty if overdue.

## C.1 Scope and Feature #6

Penalty Calculation

Automatically computes penalties based on late return (₱15 per day).

## C.1 Scope and Feature #7

Return Function

Users can return borrowed books, which removes them from the list.

## C.1 Scope and Feature #8

Settings Page

Displays user details, allows logout, and reset of the system.

## C.1 Scope and Feature #9

About Us Page

Provides information about the system and the developers.

## **D. LIMITATIONS**

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### **D.1 Limitation #1**

The system does not use a real database it only stores data in the browser

### **D.1 Limitation #2**

User accounts are fixed and cannot be created or modified

### **D.1 Limitation #3**

Data will be lost if the browser storage is cleared

### **D.1 Limitation #4**

The system cannot be accessed by multiple users at the same time

### **D.1 Limitation #5**

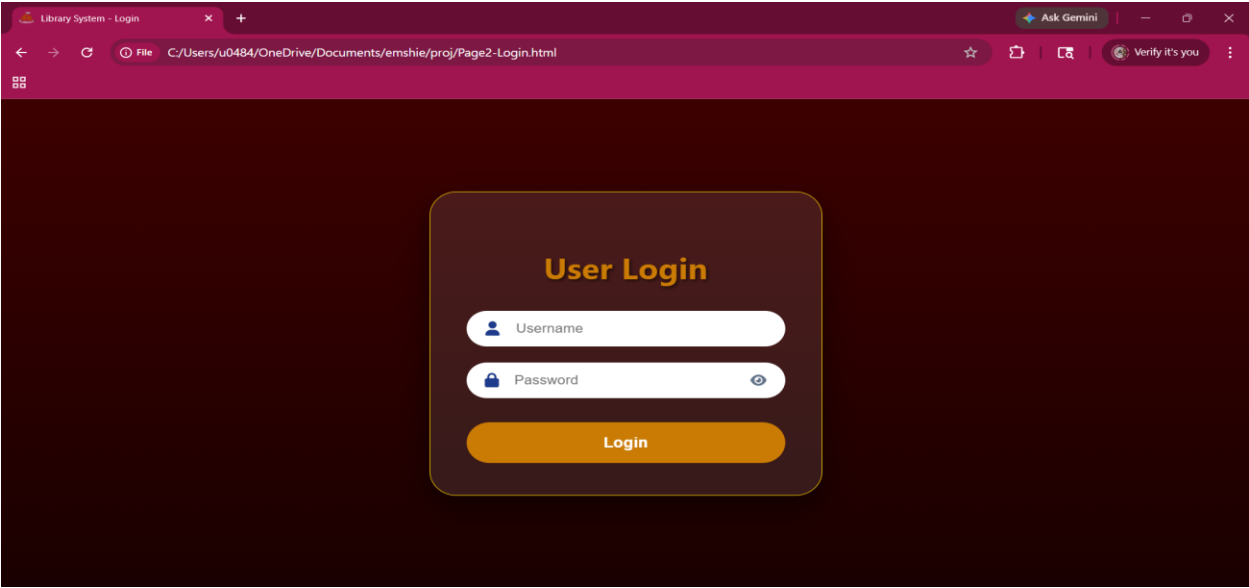
It is not connected to an actual library or real time book inventory

### **D.1 Limitation #6**

Limited security since it is only for demonstration purposes

# E. SOURCE CODES

## E.1 Login Page



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width,
initial-scale=1.0">

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

  <title>Library System - Login</title>

  <link rel="icon" type="image/png" href="libraux.png">

<style>

  /* 1. Global Page Settings */

  body {

    margin: 0;

    padding: 0;

    font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif;

    background: linear-gradient(to bottom, #3d0000,
#1a0000);

    height: 100vh;

    display: flex;

    justify-content: center;

    align-items: center;

  }

  /* 2. Main Login Box Container */

  .login-box {

    width: 350px;

    text-align: center;

    padding: 40px;

    background-color: rgba(255, 255, 255, 0.1);

    backdrop-filter: blur(15px);

    border-radius: 30px;

    border: 1px solid gold;

    box-shadow: 0px 15px 35px rgba(0,0,0,0.5);

  }

  h2 {

    margin-left: 10px;

    color: #ca7b04;

    font-size: 36px;

    margin-bottom: 30px;

    font-weight: bold;

    text-shadow: 2px 2px 4px rgba(0,0,0,0.5);

  }

  /* 3. Input Containers */

  .input-container {

    background-color: #ffffff;

    border-radius: 50px;

    margin-bottom: 20px;

    padding: 12px 20px;

    display: flex;

    align-items: center;

    transition: 0.3s;

  }

  .input-container:focus-within {

    box-shadow: 0 0 10px gold;

  }

  .input-container i {

    color: #1e3a8a;

    margin-right: 15px;

    font-size: 18px;

    width: 20px; /* Keeps icons aligned */

  }

  #togglePassword {

    cursor: pointer;

    color: #64748b;

    font-size: 16px;

    margin-right: 0;

  }

  input {

    border: none;

    background-color: transparent;

    outline: none;

    font-size: 16px;

    width: 100%;

    color: #1e3a8a;

  }

  /* 4. Login Button Styling */

  .login-btn {

    width: 100%;

    background-color: #ca7b04;

    color: white;

    border: none;

    padding: 15px;

    border-radius: 50px;

    font-size: 18px;

    font-weight: bold;

    cursor: pointer;

    transition: 0.3s;

    margin-top: 10px;

  }

  .login-btn:hover {

    background-color: orange;

    transform: translateY(-2px);

  }

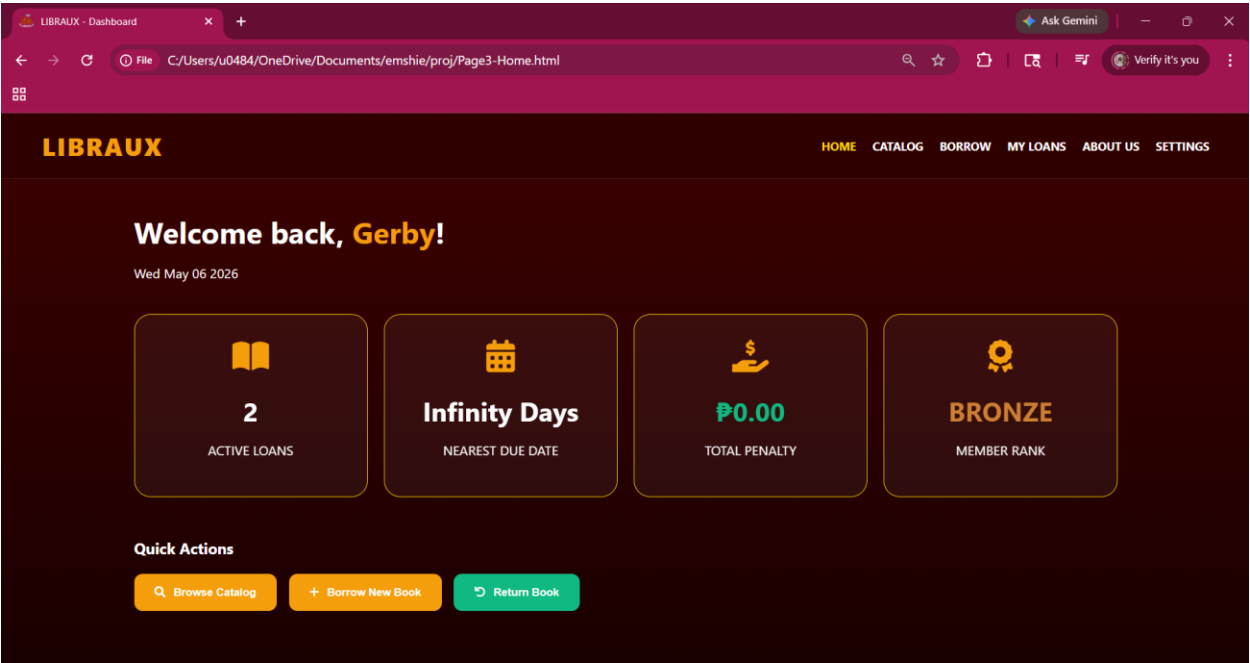
</style>

</head>

<body>
```



## E.2 Main Page



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css">

  <title>LIBRAUX - Dashboard</title>

  <link rel="icon" type="image/png" href="libraux.png">

<style>

  /* 1. Global Setup */

  body {

    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

    background: linear-gradient(to bottom, #3d0000, #1a0000);

    margin: 0;

    color: #ffffff;

    min-height: 100vh;

  }

  header {

    padding: 20px 50px;

    display: flex;

    justify-content: space-between;

    align-items: center;

    background: rgba(0, 0, 0, 0.2);

    border-bottom: 1px solid rgba(255, 255, 255, 0.1);

  }

  .logo {

    font-weight: 900;

    font-size: 30px;

    letter-spacing: 2px;

    color: #f59e0b;

  }

  nav {

    display: flex;

    gap: 20px;

  }

  nav a {

    text-decoration: none;

    color: white;

    font-size: 14px;

    font-weight: bold;

    transition: 0.3s;

  }

  nav a:hover { color: #f59e0b; }

  /* 2. Layout Wrapper */

  .dashboard-wrapper {

    padding: 40px 50px;

    max-width: 1200px;

    margin: 0 auto;

  }

  .welcome-header { margin-bottom: 40px; }

  .welcome-header h1 { font-size: 36px; margin: 0; }

  .text-accent { color: #f59e0b; }

  /* 3. Stats Grid */

  .stats-container {

    display: grid;

    grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));

    gap: 20px;

    margin-bottom: 50px;

  }

  .stat-box {

    background-color: rgba(255, 255, 255, 0.05);

    backdrop-filter: blur(10px);

    border-radius: 20px;

    border: 1px solid gold;

    padding: 30px;

    text-align: center;

    transition: 0.3s;

  }

  .stat-box i {

    font-size: 40px;

    color: #f59e0b;

    margin-bottom: 15px;

  }

  .stat-box h3 {

    font-size: 32px;

    margin: 10px 0;

  }

  /* 4. Navigation Buttons */

  .btn-group { display: flex; gap: 15px; flex-wrap: wrap; }

  .btn-main {

    background-color: #f59e0b;

    color: #ffffff;

    border: none;

    padding: 15px 25px;

    border-radius: 10px;

    cursor: pointer;

    font-weight: bold;

    transition: 0.3s;

    display: flex;

    align-items: center;

    gap: 10px;

  }

  .btn-main:hover {

    background-color: #d97706;

    transform: scale(1.05);

  }

  .btn-return { background-color: #10b981; }

  .btn-return:hover { background-color: #059669; }

  .logout-btn {

    background: none;

  }
```



```
border: 1px solid #ef4444;

color: #ef4444;

padding: 5px 15px;

border-radius: 5px;

cursor: pointer;
}

</style>

</head>

<body>

<header>

  <div class="logo">LIBRAUX</div>

  <nav>

    <a href="Page3-Home.html" style="color: gold;">HOME</a>

    <a href="Page5-Catalog.html">CATALOG</a>

    <a href="Page4-Borrow.html">BORROW</a>

    <a href="Page6-Duedate.html">MY LOANS</a>

    <a href="Page8-Aboutus.html">ABOUT US</a>

    <a href="Page7-Settings.html">SETTINGS</a>

  </nav>

</header>

<div class="dashboard-wrapper">

  <div class="welcome-header">

    <h1>Welcome back, <span id="displayUser" class="text-accent">Student</span>!</h1>

    <p id="currentDate"></p>

  </div>

  <div class="stats-container">

    <div class="stat-box">

      <i class="fas fa-book-open"></i>

      <h3 id="loanCount">0</h3>

      <p>ACTIVE LOANS</p>

    </div>

    <div class="stat-box">

      <i class="fas fa-calendar-alt"></i>

      <h3 id="daysLeft">--</h3>

      <p>NEAREST DUE DATE</p>

    </div>

    <div class="stat-box">

      <i class="fas fa-hand-holding-usd"></i>

      <h3 id="totalPenalty">P0</h3>

      <p>TOTAL PENALTY</p>

    </div>

    <div class="stat-box">

      <i class="fas fa-award"></i>

      <h3 id="memberRank">BRONZE</h3>

      <p>MEMBER RANK</p>

    </div>

  </div>

</div>
```

```
<h3>Quick Actions</h3>

<div class="btn-group">

  <button class="btn-main" onclick="location.href='Page5-Catalog.html'">

    <i class="fas fa-search"></i> Browse Catalog

  </button>

  <button class="btn-main" onclick="location.href='Page4-Borrow.html'">

    <i class="fas fa-plus"></i> Borrow New Book

  </button>

  <button class="btn-main btn-return" onclick="location.href='Page6-Duedate.html'">

    <i class="fas fa-undo"></i> Return Book

  </button>

</div>

</div>

<script>

window.onload = function() {

  // 1. Auth Check

  var user = localStorage.getItem('currentUser');

  if (!user) {

    window.location.href = "Page2-Login.html";

    return;

  }

  document.getElementById('displayUser').innerText = user;

  // 2. Set Today's Date (Midnight)

  var today = new Date();

  today.setHours(0,0,0,0);

  document.getElementById('currentDate').innerText = today.toDateString();

  // 3. Data Retrieval

  var savedData = localStorage.getItem('borrowedList');

  var borrowedList = savedData ? JSON.parse(savedData) : [];

  document.getElementById('loanCount').innerText = borrowedList.length;

  // 4. Synchronized Logic Calculations

  var shortestDays = Infinity;

  var totalPenalty = 0;

  var penaltyRatePerDay = 15; // UPDATED to 15 to match Loan Page

  borrowedList.forEach(book => {

    if (!book.returnDate) return; // Skip if no date exists

    var dueDate = new Date(book.returnDate);

    dueDate.setHours(0,0,0,0);

    // Calculate difference in time and days

    var diffTime = dueDate - today;

    var diffDays = Math.ceil(diffTime / (1000 * 60 * 60 * 24));

    // Track the most urgent book

    if (diffDays < shortestDays) {

      shortestDays = diffDays;
```

```
}

// Accumulate penalty ONLY if Today is AFTER Due Date

if (today > dueDate) {

  // This formula now matches Page 6 exactly

  var daysOverdue = Math.ceil((today - dueDate) / (1000 * 60 * 60 * 24));

  totalPenalty += (daysOverdue * penaltyRatePerDay);

}

});

// 5. Update Due Date Display

var dueDisplay = document.getElementById('daysLeft');

if (borrowedList.length > 0) {

  if (shortestDays < 0) {

    dueDisplay.innerText = "OVERDUE";

    dueDisplay.style.color = "#ef4444";

  } else if (shortestDays === 0) {

    dueDisplay.innerText = "DUE TODAY";

    dueDisplay.style.color = "#f59e0b";

  } else {

    dueDisplay.innerText = shortestDays + " Days";

    dueDisplay.style.color = "#ffffff";

  }

} else {

  dueDisplay.innerText = "None";

}

// 6. Update Penalty Display (Matches Page 6 now)

var penaltyDisplay = document.getElementById('totalPenalty');

penaltyDisplay.innerText = "P" + totalPenalty.toFixed(2);

penaltyDisplay.style.color = totalPenalty > 0 ? "#ef4444" : "#10b981";

// 7. Rank Logic

var count = borrowedList.length;

var rank = "BRONZE";

var rankColor = "#cd7f32";

if (count >= 10) { rank = "GOLD"; rankColor = "#f59e0b"; }

else if (count >= 5) { rank = "SILVER"; rankColor = "#94a3b8"; }

var rankDisplay = document.getElementById('memberRank');

rankDisplay.innerText = rank;

rankDisplay.style.color = rankColor;

};

function logout() {

  localStorage.removeItem('currentUser');

  window.location.href = "Page2-Login.html";

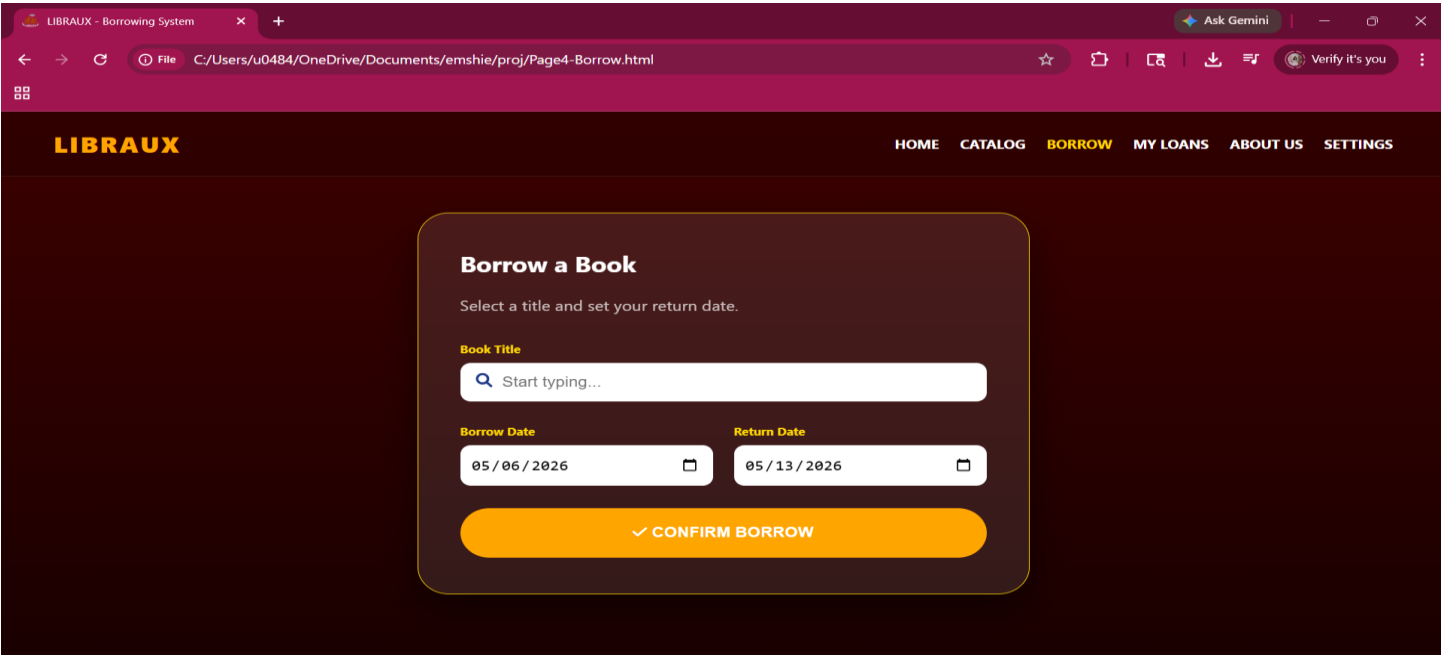
}

</script>

</body>

</html>
```

E.3 Sub page\_1 Borrow



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width,
initial-scale=1.0">

  <title>LIBRAUX - Borrowing System</title>

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

  <link rel="icon" type="image/png" href="libraux.png">

<style>

  /* 1. Global Setup */

  body {

    font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif;

    background: linear-gradient(to bottom, #3d0000,
#1a0000);

    margin: 0;

    color: white;

    min-height: 100vh;

  }

  header {

    padding: 20px 50px;

    display: flex;

    justify-content: space-between;

    align-items: center;

    background: rgba(0, 0, 0, 0.2);

    border-bottom: 1px solid rgba(255, 255, 255, 0.1);

  }

  header a {

    color: white;

    text-decoration: none;

    font-weight: bold;

    font-size: 14px;

    transition: 0.3s;

  }

  header a:hover { color: orange; }

  /* 2. Form Container */
```

```
.borrow-container {

  display: flex;

  justify-content: center;

  align-items: center;

  padding: 40px 20px;

}

.borrow-card {

  width: 100%;

  max-width: 500px;

  padding: 40px;

  background: rgba(255, 255, 255, 0.1);

  backdrop-filter: blur(15px);

  border-radius: 30px;

  border: 1px solid gold;

  box-shadow: 0px 15px 35px rgba(0,0,0,0.5);

}

.search-wrapper {

  position: relative;

  margin-bottom: 25px;

}

.search-wrapper i {

  position: absolute;

  left: 15px;

  bottom: 15px;

  color: #1e3a8a;

}

input {

  width: 100%;

  padding: 12px 12px 12px 40px;

  font-size: 16px;

  border-radius: 10px;

  border: none;

  box-sizing: border-box; /* Prevents input from
overflowing */

}

.date-row {
```

```
display: grid;

grid-template-columns: 1fr 1fr;

gap: 20px;

margin-bottom: 25px;

}

.date-row input { padding-left: 10px; }

label {

  display: block;

  margin-bottom: 8px;

  font-size: 12px;

  font-weight: bold;

  color: gold;

}

.confirmation-box {

  background: rgba(16, 185, 129, 0.2);

  border: 1px dashed #10b981;

  padding: 15px;

  border-radius: 10px;

  margin-bottom: 20px;

  display: none;

}

.btn-main {

  width: 100%;

  background-color: orange;

  color: white;

  border: none;

  padding: 18px;

  border-radius: 50px;

  font-size: 16px;

  font-weight: bold;

  cursor: pointer;

  transition: 0.3s;

}

.btn-main:hover {

  background-color: #e69500;

  transform: translateY(-2px);

}
```

```
    }

</style>

</head>

<body>

<header>

    <div class="logo" style="font-weight: 900; font-size: 24px; letter-spacing: 2px; color:orange;">LIBRAUX</div>

    <nav style="display: flex; gap: 20px;">

        <a href="Page3-Home.html">HOME</a>

        <a href="Page5-Catalog.html">CATALOG</a>

        <a href="Page4-Borrow.html" style="color: gold;">BORROW</a>

        <a href="Page6-Duedate.html">MY LOANS</a>

        <a href="Page8-Aboutus.html">ABOUT US</a>

        <a href="Page7-Settings.html">SETTINGS</a>

    </nav>

</header>

<div class="borrow-container">

    <div class="borrow-card">

        <h2 style="margin-top: 0;">Borrow a Book</h2>

        <p style="opacity: 0.7; margin-bottom: 30px;">Select a title and set your return date.</p>

        <form id="mainBorrowForm">

            <div class="search-wrapper">

                <label>Book Title</label>

                <i class="fas fa-search"></i>

                <input type="text" id="bookSearchInput" list="allBooks" placeholder="Start typing..." required oninput="showPreview()">

                <datalist id="allBooks">

                    <option value="The Hobbit"> <option value="1984"> <option value="The Alchemist">

                    <option value="Harry Potter"> <option value="Atomic Habits"> <option value="Deep Work">

                    <option value="Sapiens"> <option value="Cosmos"> <option value="Zero to One">

                    <!-- Add more options as needed -->

                </datalist>

            </div>

            <div class="confirmation-box" id="previewBox">

                <small>Selected Book:</small>

                <strong id="bookPreviewDisplay" style="display:block; color:#10b981; font-size: 18px;"></strong>

            </div>

            <div class="date-row">

                <div>
```

```
                <label>Borrow Date</label>

                <input type="date" id="bDate" required>

            </div>

            <div>

                <label>Return Date</label>

                <input type="date" id="rDate" required>

            </div>

        </div>

        <button type="submit" class="btn-main">

            <i class="fas fa-check"></i> CONFIRM BORROW

        </button>

    </form>

</div>

<script>

    window.onload = function() {

        // 1. Set default Borrow Date to Today

        const today = new Date();

        const todayStr = today.toISOString().split('T')[0];

        document.getElementById('bDate').value = todayStr;

        // 2. Set default Return Date to 7 days from now (Standard Policy)

        const nextWeek = new Date();

        nextWeek.setDate(today.getDate() + 7);

        document.getElementById('rDate').value = nextWeek.toISOString().split('T')[0];

        // 3. Handle Catalog Redirection

        const selected = localStorage.getItem('selectedBook');

        if(selected) {

            document.getElementById('bookSearchInput').value = selected;

            showPreview();

            localStorage.removeItem('selectedBook');

        }

    };

    function showPreview() {

        const val = document.getElementById('bookSearchInput').value;

        const box = document.getElementById('previewBox');

        const display = document.getElementById('bookPreviewDisplay');

        if(val.trim().length > 0) {

            box.style.display = "block";

            display.innerText = val;

        } else {

            box.style.display = "none";

        }

    }

</script>
```

```
    }

    document.getElementById('mainBorrowForm').onsubmit = function(e) {

        e.preventDefault();

        const bookTitle = document.getElementById('bookSearchInput').value;

        const startDate = document.getElementById('bDate').value;

        const endDate = document.getElementById('rDate').value;

        // Date Validation

        if (endDate < startDate) {

            alert("Error: Return date cannot be before borrow date!");

            return;

        }

        // The Object structure that all other pages rely on

        const newLoan = {

            title: bookTitle,

            borrowDate: startDate,

            returnDate: endDate,

            penaltyRate: 15 // Saving the rate inside the object ensures consistency

        };

        let borrowedList = JSON.parse(localStorage.getItem('borrowedList') || "[]");

        // Check for duplicates

        const isAlreadyBorrowed = borrowedList.some(item => item.title === bookTitle);

        if (isAlreadyBorrowed) {

            alert("You are already borrowing this book!");

        } else {

            borrowedList.push(newLoan);

            localStorage.setItem('borrowedList', JSON.stringify(borrowedList));

            alert(` Successfully Borrowed: ${bookTitle}`);

            // Redirect to My Loans to confirm the data is there

            window.location.href = "Page6-Duedate.html";

        }

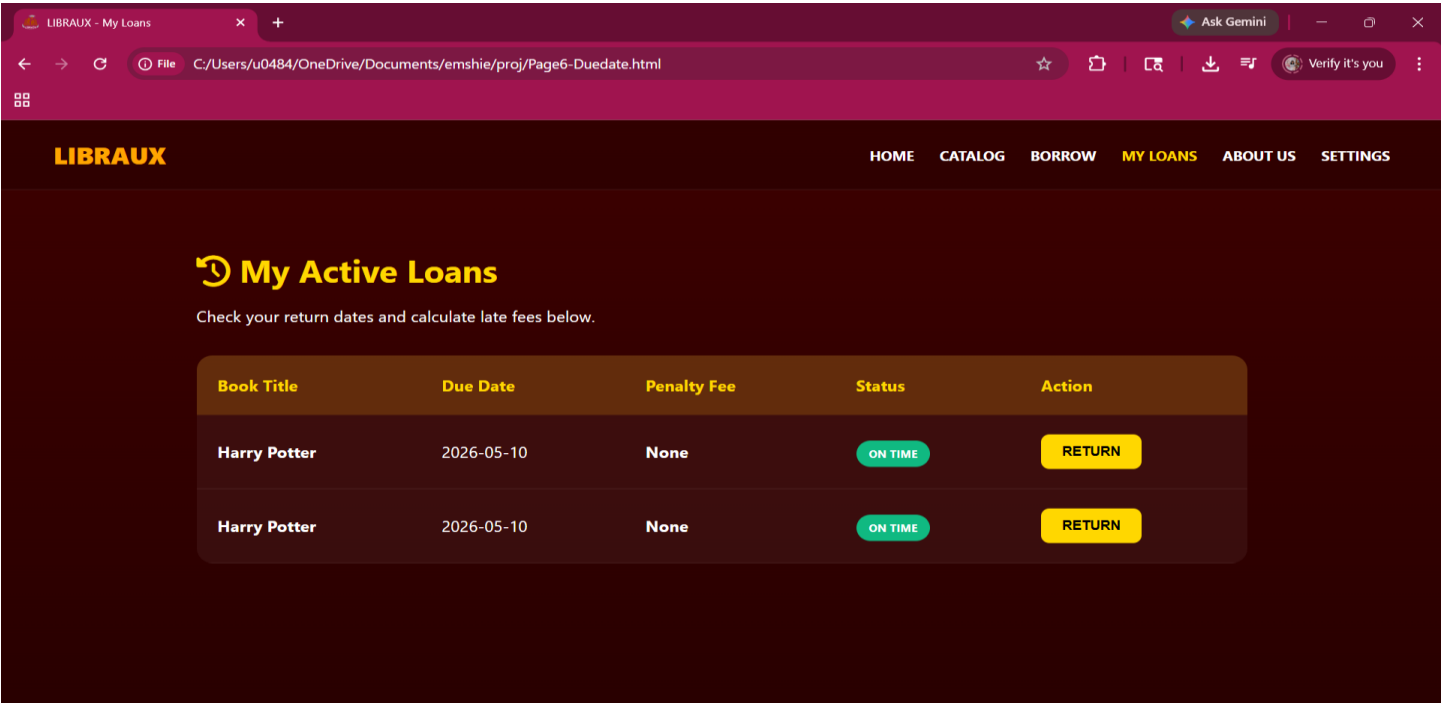
    };

</script>

</body>

</html>
```

E.4 Sub page\_2



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>LIBRAUX - My Loans</title>

  <link rel="icon" type="image/png" href="libraux.png">

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

  <style>

    body {

      background-image: linear-gradient(to bottom, rgb(61,
0, 0), rgb(41, 0, 0));

      font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif;

      color: white;

      margin: 0;

      min-height: 100vh;

    }

    header {

      padding: 20px 50px;

      display: flex;

      justify-content: space-between;

      align-items: center;

      background: rgba(0, 0, 0, 0.2);

      border-bottom: 1px solid rgba(255, 255, 255, 0.1);

    }

    header a {

      color: white;

      text-decoration: none;

      font-weight: bold;

      font-size: 14px;

      margin-left: 20px;

    }

    header a:hover { color: orange; }

    .loans-container { padding: 40px 50px; max-width:
1000px; margin: 0 auto; }

    h1 { color: gold; margin-bottom: 10px; }

    .loan-table {

      width: 100%;

      border-collapse: collapse;

      margin-top: 30px;

      background: rgba(255, 255, 255, 0.05);

      border-radius: 15px;

      overflow: hidden;

    }

    .loan-table th { background: rgba(153, 91, 11, 0.4);
padding: 20px; text-align: left; color: gold; }

    .loan-table td { padding: 20px; border-bottom: 1px solid
rgba(255, 255, 255, 0.1); }

    .status-badge { padding: 6px 12px; border-radius: 20px;
font-size: 11px; font-weight: bold; }

    .status-on-time { background: #10b981; color: white; }

    .status-overdue { background: #ef4444; color: white; }

    .penalty-text { color: #ff9999; font-weight: bold; }

    .return-btn {

      background: gold;

      color: black;

      border: none;

      padding: 10px 20px;

      border-radius: 8px;

      cursor: pointer;

      font-weight: bold;

    }

    .empty-state { text-align: center; padding: 100px;
opacity: 0.5; border: 2px dashed rgba(255,255,255,0.1); }

</style>

</head>

<body>

  <header>

    <div class="logo" style="font-weight: 900; font-size:
24px; color:orange;">LIBRAUX</div>

    <nav>

      <a href="Page3-Home.html">HOME</a>

      <a href="Page5-Catalog.html">CATALOG</a>

      <a href="Page4-Borrow.html">BORROW</a>

      <a href="Page6-Duedate.html" style="color:
gold;">MY LOANS</a>

      <a href="Page8-Aboutus.html">ABOUT US</a>

      <a href="Page7-Settings.html">SETTINGS</a>

    </nav>

  </header>

  <div class="loans-container">

    <h1><i class="fas fa-history"></i> My Active
Loans</h1>

    <p>Check your return dates and calculate late fees
below.</p>

    <div id="loanContent">

      <table class="loan-table" id="loanTable">

        <thead>

          <tr>

            <th>Book Title</th>

            <th>Due Date</th>

            <th>Penalty Fee</th>

            <th>Status</th>

            <th>Action</th>

          </tr>

        </thead>

        <tbody id="loanBody"></tbody>

      </table>

      <div id="noLoans" class="empty-state"
style="display: none;">

        <i class="fas fa-book-reader fa-3x"></i>

        <p>No active loans. Keep on reading!</p>

      </div>

    </div>

  </div>

  <script>

    function loadLoans() {

      const borrowedList =
JSON.parse(localStorage.getItem('borrowedList') || "[]");
```

```
const table = document.getElementById('loanTable');

const body = document.getElementById('loanBody');

const emptyState = document.getElementById('noLoans');

if (borrowedList.length === 0) {

    table.style.display = "none";

    emptyState.style.display = "block";

    return;

}

table.style.display = "table";

emptyState.style.display = "none";

body.innerHTML = "";

// 1. Get Today's Date at Midnight

const today = new Date();

today.setHours(0, 0, 0, 0);

borrowedList.forEach((book, index) => {

    // 2. Correctly extract data

    let title = (typeof book === 'object') ? book.title : book;

    // Use the saved date, or a past date if you want to
    FORCE a penalty for testing

    let dueDateStr = (typeof book === 'object' &&
    book.returnDate) ? book.returnDate : "2026-05-10";

    // 3. Robust Date Comparison

    const dueDate = new Date(dueDateStr);
```

```
// We compare milliseconds to ensure accuracy

// checks if the book is overdue

const isOverdue = dueDate.getTime() <
today.getTime();

let penaltyDisplay = "None";

let statusClass = "status-on-time";

let statusText = "ON TIME";

if (isOverdue) {

    // Calculate days late

    const diffTime = today.getTime() - dueDate.getTime();

    const diffDays = Math.ceil(diffTime / (1000 * 60 * 60
    * 24)); // converts

    const dailyFine = 15.00;

    const totalFine = diffDays * dailyFine;

    penaltyDisplay = `P${totalFine.toFixed(2)}`;

    statusClass = "status-overdue";

    statusText = "OVERDUE";

}

// 4. Update the Row

const row = document.createElement('tr');

//penalty

row.innerHTML = `

    <td><strong>${title}</strong></td>
```

```
<td>${dueDateStr}</td>

    <td class="penalty-text" style="color: ${isOverdue ?
    '#ff9999' : '#fff'}">${penaltyDisplay}</td>

    <td><span class="status-badge
    ${statusClass}">${statusText}</span></td>

    <td><button class="return-btn"
    onclick="returnBook(${index})">RETURN</button></td>

`;

    body.appendChild(row);

});

}

function returnBook(index) {

    let borrowedList =
    JSON.parse(localStorage.getItem('borrowedList') || "[]");

    const bookTitle = typeof borrowedList[index] === 'object'
    ? borrowedList[index].title : borrowedList[index];

    borrowedList.splice(index, 1);

    localStorage.setItem('borrowedList',
    JSON.stringify(borrowedList));

    alert(`Successfully returned: ${bookTitle}`);

    loadLoans();

}

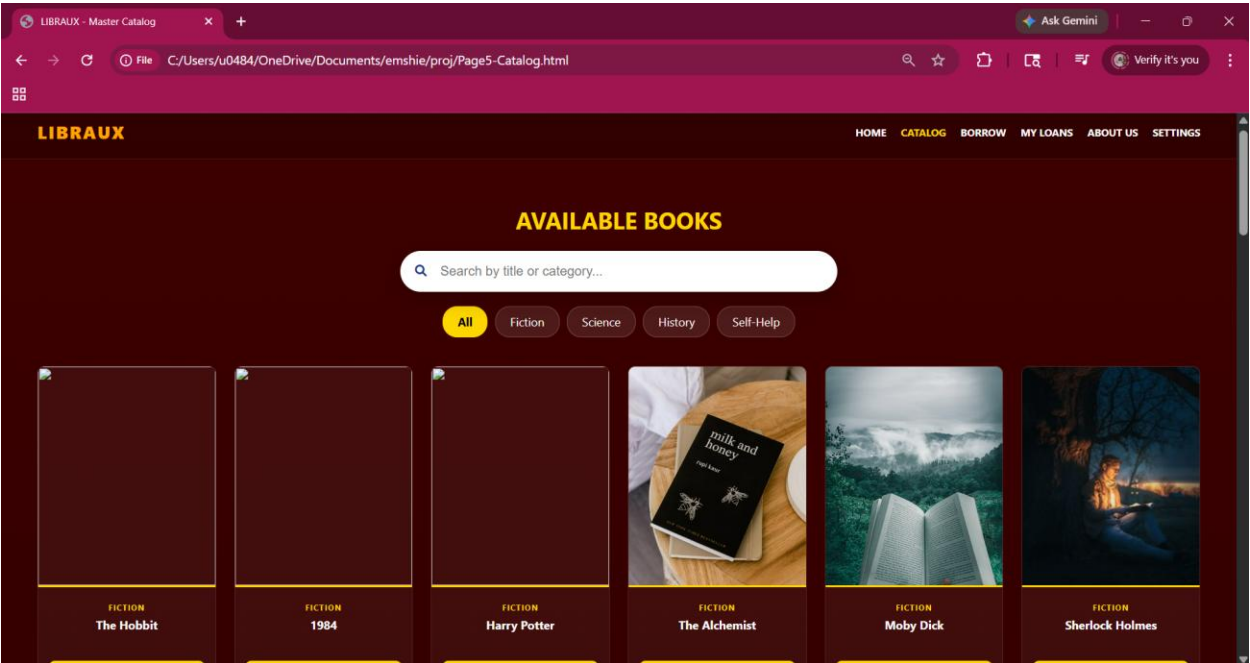
window.onload = loadLoans;

</script>

</body>

</html>
```

## E.5 Catalog page



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>LIBRAUX - Master Catalog</title>

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

  <style>

    /* --- GLOBAL SETUP --- */

    body {

      background: linear-gradient(to bottom, #3d0000,
#1a0000);

      font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif;

      color: white;

      margin: 0;

      min-height: 100vh;

    }

    header {

      padding: 20px 50px;

      display: flex;

      justify-content: space-between;

      align-items: center;

      background: rgba(0, 0, 0, 0.2);

      border-bottom: 1px solid rgba(255, 255, 255, 0.1);

    }

    header a {

      color: white;

      text-decoration: none;

      font-weight: bold;

      font-size: 14px;

      transition: 0.3s;

    }

    header a:hover { color: orange; }

    /* --- SEARCH & FILTER SECTION --- */

    .catalog-container { padding: 40px 50px; }

    .search-section {

      text-align: center;

      margin-bottom: 40px;

    }

    h1 {

      color: gold;

      border: 10px;

    }

    .search-box {

      position: relative;

      max-width: 600px;

      margin: 0 auto 20px;

    }

    .search-box i {

      position: absolute;

      left: 20px; top: 50%;

      transform: translateY(-50%);

      color: #1e3a8a;

    }

    #catalogSearch {

      width: 100%;

      padding: 18px 55px;

      border-radius: 50px;

      border: none;

      font-size: 18px;

      box-shadow: 0 10px 30px rgba(0,0,0,0.2);

      outline: none;

      box-sizing: border-box;

    }

    /* --- CATEGORY TABS --- */

    .category-tabs {

      display: flex;

      justify-content: center;

      gap: 10px;

      margin-bottom: 30px;

    }

    .flex-wrap: wrap;

  }

  .tab {

    padding: 10px 20px;

    border-radius: 20px;

    background: rgba(255,255,255,0.1);

    cursor: pointer;

    border: 1px solid rgba(255,255,255,0.2);

    transition: 0.3s;

    color: white;

  }

  .tab.active {

    background: gold;

    color: black;

    font-weight: bold;

    border-color: gold;

  }

  /* --- BOOK GRID --- */

  .book-grid {

    display: grid;

    grid-template-columns: repeat(auto-fill,
minmax(220px, 1fr));

    gap: 25px;

  }

  .book-card {

    display: flex;

    flex-direction: column;

    transition: 0.3s;

    border: 1px solid rgba(255,255,255,0.1);

    background: rgba(255,255,255,0.05);

    border-radius: 10px;

    overflow: hidden;

  }

  .book-card:hover {

    transform: translateY(-10px);

    border-color: gold;

  }
```

```

}

.book-img {

    width: 100%;

    height: 300px;

    object-fit: cover;

    border-bottom: 3px solid gold;
}

.book-info {

    padding: 15px;

    text-align: center;

    flex-grow: 1;
}

.book-info h4 {

    font-size: 16px;

    margin: 5px 0;

    height: 40px;

    overflow: hidden;
}

.book-info span {

    font-size: 11px;

    color: gold;

    font-weight: bold;

    text-transform: uppercase;

    letter-spacing: 1px;
}

.btn-borrow {

    margin: 0 15px 15px;

    padding: 12px;

    border: none;

    border-radius: 8px;

    background: gold;

    color: black;

    font-weight: 900;

    cursor: pointer;

    transition: 0.3s;
}

.btn-borrow:hover { background: yellow; }

</style>

</head>

<body>

<header>

    <div class="logo" style="font-weight: 900; font-size: 24px; letter-spacing: 2px; color:orange;">LIBRAUX</div>

    <nav style="display: flex; gap: 20px;">

        <a href="Page3-Home.html">HOME</a>

        <a href="Page5-Catalog.html" style="color: gold;">CATALOG</a>

        <a href="Page4-Borrow.html">BORROW</a>

        <a href="Page6-Duedate.html">MY LOANS</a>

        <a href="Page8-Aboutus.html">ABOUT US</a>

        <a href="Page7-Settings.html">SETTINGS</a>

    </nav>

</header>

<div class="catalog-container">

    <div class="search-section">

        <h1>AVAILABLE BOOKS</h1>

        <div class="search-box">

            <i class="fas fa-search"></i>

            <input type="text" id="catalogSearch" placeholder="Search by title or category..." onkeyup="filterBooks()">

        </div>

        <div class="category-tabs" id="tabs">

            <div class="tab active" onclick="filterTab('All', this)">All</div>

            <div class="tab" onclick="filterTab('Fiction', this)">Fiction</div>

            <div class="tab" onclick="filterTab('Science', this)">Science</div>

            <div class="tab" onclick="filterTab('History', this)">History</div>

            <div class="tab" onclick="filterTab('Self-Help', this)">Self-Help</div>

        </div>

        <div class="book-grid" id="bookGrid"></div>

    </div>

<script>

    const books = [

        // Fiction (10)

        { t: "The Hobbit", c: "Fiction", i: "https://images.unsplash.com/photo-1629992101753-56d196c8aced?w=400" },

        { t: "1984", c: "Fiction", i: "https://images.unsplash.com/photo-1543004271-92523286eb29?w=400" },

        { t: "Harry Potter", c: "Fiction", i: "https://images.unsplash.com/photo-1610466024868-37209772a15c?w=400" },

        { t: "The Alchemist", c: "Fiction", i: "https://images.unsplash.com/photo-1544947950-fa07a98d237f?w=400" },

        { t: "Moby Dick", c: "Fiction", i: "https://images.unsplash.com/photo-1541963463532-d68292c34b19?w=400" },

        { t: "Sherlock Holmes", c: "Fiction", i: "https://images.unsplash.com/photo-1587876931567-564ce588bfbd?w=400" },

        { t: "The Night Circus", c: "Fiction", i: "https://images.unsplash.com/photo-1512820790803-83ca734da794?w=400" },

        { t: "Life of Pi", c: "Fiction", i: "https://images.unsplash.com/photo-1474932430478-3a7fb9082db4?w=400" },

        { t: "The Little Prince", c: "Fiction", i: "https://images.unsplash.com/photo-1512591046661-863fad28590d?w=400" },

        { t: "Frankenstein", c: "Fiction", i: "https://images.unsplash.com/photo-1589829085413-56de8ae18c73?w=400" },

        // Science (10)

        { t: "Cosmos", c: "Science", i: "https://images.unsplash.com/photo-1464802686167-b939a6910659?w=400" },

        { t: "Brief History of Time", c: "Science", i: "https://images.unsplash.com/photo-1451187580459-43490279c0fa?w=400" },

        { t: "The Gene", c: "Science", i: "https://images.unsplash.com/photo-1530026405186-ed1f139313f8?w=400" },

        { t: "Quantum Physics", c: "Science", i: "https://images.unsplash.com/photo-1635070041078-e363dbe005cb?w=400" },

        { t: "Chemistry Essentials", c: "Science", i: "https://images.unsplash.com/photo-1532187863486-abf9d39d6d2d?w=400" },

        { t: "Biology 101", c: "Science", i: "https://images.unsplash.com/photo-1576086213369-97a306d36557?w=400" },

        { t: "Silent Spring", c: "Science", i: "https://images.unsplash.com/photo-1501166073351-be743d5a1d3a?w=400" },

        { t: "Astrophysics", c: "Science", i: "https://images.unsplash.com/photo-1506318137071-a8e063b4b4bf?w=400" },

        { t: "The Selfish Gene", c: "Science", i: "https://images.unsplash.com/photo-1471017851983-fc49d89c57c2?w=400" },

        { t: "Neuroscience", c: "Science", i: "https://images.unsplash.com/photo-1559757175-5700dde675bc?w=400" },

        // History (10)

        { t: "Sapiens", c: "History", i: "https://images.unsplash.com/photo-1512820790803-83ca734da794?w=400" },

        { t: "Ancient Rome", c: "History", i: "https://images.unsplash.com/photo-1552832230-c0197dd311b5?w=400" },

        { t: "The Silk Roads", c: "History", i: "https://images.unsplash.com/photo-1526367790999-0150786486a2?w=400" },

        { t: "Guns Germs Steel", c: "History", i: "https://images.unsplash.com/photo-1521587760476-6c12a4b040da?w=400" },

        { t: "Napoleon", c: "History", i: "https://images.unsplash.com/photo-1599409675172-03a9608898cc?w=400" },

        { t: "Leonardo Da Vinci", c: "History", i: "https://images.unsplash.com/photo-1579783902614-a3fb3927b6a5?w=400" },

        { t: "The Vikings", c: "History", i: "https://images.unsplash.com/photo-1534073828943-f801091bb18c?w=400" },

        { t: "WWII: The Blitz", c: "History", i: "https://images.unsplash.com/photo-1513569771920-c9e1d31714af?w=400" },

        { t: "Ancient Egypt", c: "History", i: "https://images.unsplash.com/photo-1503177119275-0aa32b3a9368?w=400" },

        { t: "The Crusades", c: "History", i: "https://images.unsplash.com/photo-1461360228754-6e81c478c882?w=400" },

        // Self-Help (10)

        { t: "Atomic Habits", c: "Self-Help", i: "https://images.unsplash.com/photo-1589998059171-988d687df646?w=400" },

        { t: "Deep Work", c: "Self-Help", i: "https://images.unsplash.com/photo-1499750310107-5fef28a66643?w=400" },

        { t: "Thinking Fast and Slow", c: "Self-Help", i: "https://images.unsplash.com/photo-1456513080510-7bf3a84b82f8?w=400" },

        { t: "Educated", c: "Self-Help", i: "https://images.unsplash.com/photo-1512820790803-83ca734da794?w=400" },

        { t: "Becoming", c: "Self-Help", i: "https://images.unsplash.com/photo-1541963463532-d68292c34b19?w=400" },

        { t: "Quiet", c: "Self-Help", i: "https://images.unsplash.com/photo-1492538368677-f6e0afe31dcc?w=400" },

        { t: "Grit", c: "Self-Help", i: "https://images.unsplash.com/photo-1506784365847-bbad939e9335?w=400" },

        { t: "The 5 AM Club", c: "Self-Help", i: "https://images.unsplash.com/photo-1495360010541-f48722b34fd?w=400" },

        { t: "Zero to One", c: "Self-Help", i: "https://images.unsplash.com/photo-1512428559083-5d0df9a30143?w=400" },

        { t: "Limitless", c: "Self-Help", i: "https://images.unsplash.com/photo-1481627834876-b7833e8f5570?w=400" }

    ];

    const grid = document.getElementById('bookGrid');

    function display(data) {

        grid.innerHTML = '';

        data.forEach(book => {

            const card = document.createElement('div');

            card.className = "book-card";

            card.innerHTML = `

                <div class="book-info">
            `
        })
    }
}

```

```

        <span>${book.c}</span>

        <h4>${book.t}</h4>

    </div>

    <button class="btn-borrow"
onclick="startBorrow('${book.t}')">BORROW</button>

    `;

    grid.appendChild(card);

    });

}

// SEARCH FILTER

function filterBooks() {

    const query =
document.getElementById('catalogSearch').value.toLowerCa
se();
```

```

        const filtered = books.filter(b =>
b.t.toLowerCase().includes(query) ||
b.c.toLowerCase().includes(query));

        display(filtered);

    }

    // TAB FILTER

    function filterTab(cat, el) {

        document.querySelectorAll('.tab').forEach(t =>
t.classList.remove('active'));

        el.classList.add('active');

        if(cat === 'All') display(books);

        else display(books.filter(b => b.c === cat));

    }
```

```

// BRIDGE TO PAGE 4

function startBorrow(title) {

    localStorage.setItem('selectedBook', title);

    window.location.href = "Page4-Borrow.html";

}

window.onload = () => display(books);

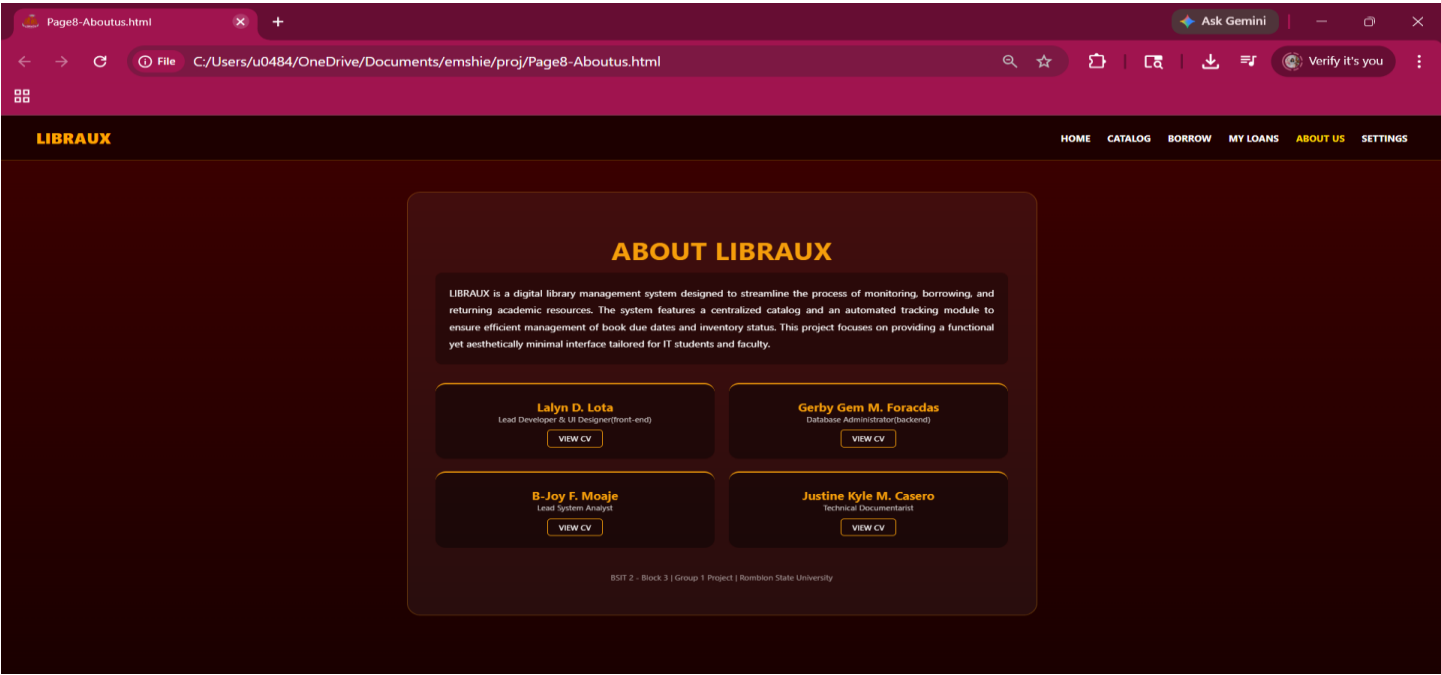
</script>

</body>

</html>
```



## E.2 About Us Page



```
<link rel="icon" type="image/png" href="libraux.png">

<style>

/* --- FIXED YOUR HEADER --- */

header {

padding: 20px 50px;

display: flex;

justify-content: space-between;

align-items: center;

/* Fixed to match your body's deep red theme */

background: rgba(26, 0, 0, 0.9);

border-bottom: 1px solid rgba(245, 158, 11, 0.3);

position: sticky;

top: 0;

z-index: 1000;

}

header a {

color: white;

text-decoration: none;

font-weight: bold;

font-size: 14px;

margin-left: 20px;

transition: 0.3s;

}

header a:hover {

color: #f59e0b;

}

/* --- YOUR ORIGINAL BODY DESIGN --- */

body {

background: linear-gradient(to bottom, #3d0000, #1a0000);

color: white;

font-family: 'Segoe UI', sans-serif;

margin: 0;

padding: 0px 0px 50px 0px; /* Adjusted to prevent header overlap */

}

.about-container {

max-width: 900px;

margin: 50px auto;

text-align: center;

}

.glass-card {

background: rgba(255, 255, 255, 0.05);

backdrop-filter: blur(10px);

border: 1px solid rgba(245, 158, 11, 0.3);

border-radius: 20px;

padding: 40px;

}

h1 { color: #f59e0b; font-size: 40px; margin-bottom: 10px; }

/* --- ADDED DESCRIPTION STYLING --- */

.project-description {

text-align: justify;

line-height: 1.8;

margin-bottom: 30px;

padding: 20px;

background: rgba(0, 0, 0, 0.3);

border-radius: 10px;

font-size: 15px;

}

.team-grid {

display: grid;

grid-template-columns: repeat(2, 1fr);

gap: 20px;

}

.member-card {

background: rgba(0,0,0,0.4);

padding: 25px;

border-radius: 15px;

border-top: 3px solid #f59e0b;

transition: 0.3s;

}

.member-card:hover { transform: translateY(-5px); background: rgba(0,0,0,0.6); }

.name { display: block; font-weight: bold; color: #f59e0b; font-size: 18px; }

.role { display: block; font-size: 12px; margin-bottom: 15px; opacity: 0.7; }

.cv-btn {

text-decoration: none;

color: white;

background: transparent;

border: 1px solid #f59e0b;

padding: 5px 15px;

border-radius: 5px;

font-size: 12px;

transition: 0.3s;

}

.cv-btn:hover { background: #f59e0b; color: black; }

</style>

<header>

<div class="logo" style="font-weight: 900; font-size: 24px; color:orange;">LIBRAUX</div>

<nav>

<a href="Page3-Home.html">HOME</a>

<a href="Page5-Catalog.html">CATALOG</a>

<a href="Page4-Borrow.html">BORROW</a>

<a href="Page6-Duedate.html">MY LOANS</a>

<a href="Page8-Aboutus.html" style="color:gold;">ABOUT US</a>

<a href="Page7-Settings.html">SETTINGS</a>

</nav>

</header>

<div class="about-container">

<div class="glass-card">

<h1>ABOUT LIBRAUX</h1>

<!-- Added Project Description -->

<div class="project-description">

LIBRAUX is a digital library management system designed to streamline the process of monitoring,
```

borrowing, and returning academic resources. The system features a centralized catalog and an

automated tracking module to ensure efficient management of book due dates and inventory status.

This project focuses on providing a functional yet aesthetically minimal interface tailored for

IT students and faculty.

</div>

<div class="team-grid">

<div class="member-card">

<span class="name">Lalyn D. Lota</span>

<span class="role">Lead Developer & UI Designer(front-end)</span>

<a href="lalyn\_Cv.html" class="cv-btn">VIEW CV</a>

</div>

<div class="member-card">

<span class="name">Gerby Gem M. Foracdas</span>

<span class="role">Database Administrator(backend)</span>

<a href="gerb\_cv.html" class="cv-btn">VIEW CV</a>

</div>

<div class="member-card">

<span class="name">B-Joy F. Moaje</span>

<span class="role">Lead System Analyst</span>

<a href="joy\_cv.html" class="cv-btn">VIEW CV</a>

</div>

<div class="member-card">

<span class="name">Justine Kyle M. Casero</span>

<span class="role">Technical Documentarist</span>

<a href="resume.html" class="cv-btn">VIEW CV</a>

</div>

</div>

<p style="margin-top: 40px; font-size: 12px; opacity: 0.5;">

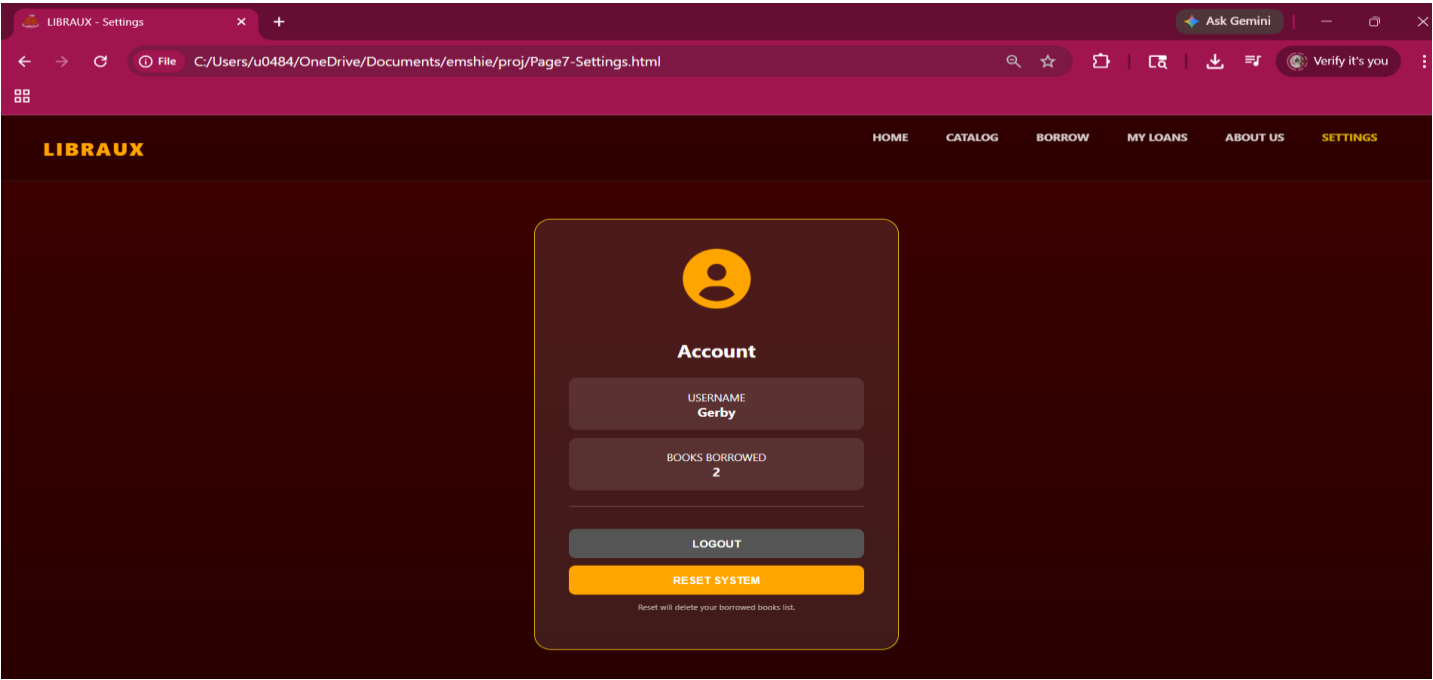
BSIT 2 - Block 3 | Group 1 Project | Romblon State University

</p>

</div>

</div>

## E.2 Settings Page



```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>LIBRAUX - Settings</title>

  <link rel="icon" type="image/png"
href="libraux.png">

  <link rel="stylesheet" href="External.css">

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

<style>

  header {

    padding: 20px 50px;

    display: flex;

    justify-content: space-between;

    align-items: center;

    background: rgba(0, 0, 0, 0.2);

    border-bottom: 1px solid rgba(255, 255,
255, 0.1);

  }

  header a {

    color: white;

    text-decoration: none;

    font-weight: bold;

    font-size: 14px;

    margin-left: 20px;

  }

  header a:hover { color: orange; }

  body { background-image: linear-gradient(to
bottom, rgb(61, 0, 0), rgb(41, 0, 0)); color: white; }

  .settings-wrapper { padding: 50px 20px;
display: flex; justify-content: center; }

  .settings-card { width: 350px; padding: 40px;
text-align: center; background:
rgba(255,255,255,0.1); border-radius: 20px;
border: 1px solid gold; }

  .profile-icon { font-size: 80px; color: orange;
margin-bottom: 20px; }

  .user-detail { background:
rgba(255,255,255,0.1); padding: 15px; border-
radius: 10px; margin: 10px 0; }

  .btn { width: 100%; padding: 12px; border-
radius: 8px; border: none; font-weight: bold;
cursor: pointer; margin-top: 10px; color: white; }

  .btn-logout { background: #555; }

  .btn-reset { background: orange; }

</style>

</head>

<body>

  <header>

    <div class="logo" style="font-weight: 900;
font-size: 24px; color:orange;">LIBRAUX</div>

    <nav>

      <a href="Page3-Home.html">HOME</a>

      <a href="Page5-
Catalog.html">CATALOG</a>

      <a href="Page4-
Borrow.html">BORROW</a>

      <a href="Page6-Duedate.html" >MY
LOANS</a>

      <a href="Page8-Aboutus.html">ABOUT
US</a>

      <a href="Page7-Settings.html"
style="color: gold;">SETTINGS</a>

    </nav>

  </header>

  <div class="settings-wrapper">

    <div class="settings-card">

      <i class="fas fa-user-circle profile-
icon"></i>

      <h2>Account</h2>

      <div class="user-detail">

        <small>USERNAME</small><br>

        <strong id="stName">---</strong>

      </div>

      <div class="user-detail">

        <small>BOOKS
BORROWED</small><br>

        <strong id="stLoans">0</strong>

      </div>

      <div>

        <button class="btn btn-logout"
onclick="goLogout()">LOGOUT</button>

        <!-- LOGOUT: Just leaves the account -->

        <button class="btn btn-reset"
onclick="goReset()">RESET SYSTEM</button>

        <!-- RESET: Deletes all books -->

        <p style="font-size: 10px; opacity: 0.6;
margin-top: 10px;">

          Reset will delete your borrowed books
list.

        </p>

      </div>

    </div>

  </div>

<script>

  window.onload = function() {

    // 1. Show the Username

    var name =
localStorage.getItem('currentUser');

    if (name == null) {

      document.getElementById('stName').innerText =
"Guest";

    } else {

      document.getElementById('stName').innerText =
name;

    }

    // 2. Count the books

    var data =
localStorage.getItem('borrowedList');

    if (data != null) {

      var list = JSON.parse(data);

      document.getElementById('stLoans').innerText =
list.length;

    }

  }

}
```

```
    }  
};  
  
// Function for Logout (Keeps the books in  
memory)  
  
function goLogout() {  
    var answer = confirm("Are you sure you  
want to logout?");  
    if (answer == true) {  
        localStorage.removeItem('currentUser');
```

```
        window.location.href = "Page2-  
Login.html";  
    }  
}  
  
// Function for Reset (Deletes the books too)  
  
function goReset() {  
    var answer = confirm("Warning: This  
deletes all books! Continue?");  
    if (answer == true) {
```

```
        localStorage.clear();  
        window.location.href = "Page1-  
Cover.html";  
    }  
}  
</script>  
</body>  
</html>
```

## F. CODE WRITING

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| Part of the code            | Created by:        |
|-----------------------------|--------------------|
| 1. Login Logic & Navigation | B-Joy Moaje        |
| 2. Technical Lead           | Gerby Gem Forcadas |
| 3. Dashboard UI & Layout    | Lalyn Lota         |
| 4. Testing & Documentation  | Justin Casero      |

## G. GROUP MEMBERS

---



Moaje, B-Joy F.  
Tumingad, Odiongan, Romblon  
BSIT 2 – Block 3  
jymoaje@gmail.com  
Despite the challenges with bugs and teamwork,  
we improved our HTML, CSS, and JavaScript skills  
and learned patience and collaboration.



Lota, Lalyn D.  
Liwayway, Odiongan, Romblon  
BSIT 2 – Block 3  
Lalynlota18@gmail.com  
Give us a clearer idea of how web systems are  
designed and developed from start to finish.



Casero, Justine Kyle M.  
Liwanag, Odiongan, Romblon  
BSIT 2 – Block 3  
Justinecasero2006@gmail.com  
This project improved our skills in  
HTML, CSS, and JavaScript, as well as  
our ability to work as a 4 members or



Forcadas, Gerby Gem M.  
Batiano, Odiongan, Romblon  
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gemforcadas@gmail.com  
Working this project is a great help  
to improve more of our knowledge in  
creating a website.

## **H. ACKNOWLEDGEMENT**

---

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